

RAHBAR

Comprehensive Public Transit & Personal Safety Ecosystem

1. Executive Summary

Rahbar has evolved from a transit security system into a comprehensive, dual-purpose safety ecosystem. Designed for Pakistan's specific security challenges, Rahbar protects users during everyday commutes—whether walking, commuting to university, returning from work, or traveling inter-city on public transit. It empowers users against harassment, theft, and missing person scenarios by combining automated emergency protocols, digital evidence collection, offline location tracking, and law enforcement connectivity into a single intelligent platform.

2. Dual Problem Statement

- **Personal Vulnerability on Streets:** Individuals facing street harassment, stalking, or mugging lack an instant, automated way to trigger a comprehensive police/guardian response without alerting their attacker. Furthermore, loss of internet connectivity often leaves them completely untraceable.
- **Transit Security Loopholes:** Public transport lacks real-time passenger safety tracking and automated background checks, making check-posts inefficient and leaving passengers at risk when devices power down.

3. Core System Modules

Module 1: Personal SOS & Automated Emergency Protocol

A multi-layered defense mechanism for personal safety. If a user feels threatened (e.g., being followed), triggering the SOS initiates an **Automated Phone Call** to the police or guardians utilizing a pre-recorded emergency voice message. If the call fails, the system automatically falls back to an **SMS Alert** containing the live location sent to a designated emergency helpline or contacts.

Module 2: Evidence Collection & WhatsApp Automation

The system allows users to discreetly activate audio or video recording to capture evidence of harassment or theft. This media is securely saved in a local storage folder. The app integrates with WhatsApp (via Intent APIs) to instantly transmit the user's situation and media evidence to predefined emergency contacts.

Module 3: Offline "Last Known Location" Breadcrumb Trail

To combat internet outages and missing person scenarios, the app pings the GPS every 10 seconds, saving coordinates directly to a local device folder. If the user loses connection or goes missing, this log acts as a critical evidence trail. Once connection is restored (or via SMS trigger), the system instantly transmits the **last visited area/saved location** to the police and guardians to drastically narrow down the search radius.

Module 4: Smart Transit Integration

When boarding inter-city transport, passengers register via CNIC. The system cross-references a (mock) criminal database, issuing silent alerts to nearby police if a flagged individual boards. Cleared buses receive a "Green Signal" to bypass manual check-posts, optimizing travel while maintaining background GPS and battery-low alerts to guardians.

4. Technical Architecture & Viva Defense Strategies

Rahbar utilizes a robust tech stack (Node.js/Python backend, MySQL/PostgreSQL, Google Maps/SMS APIs). For the defense, key technical justifications include:

- **WhatsApp Automation:** Leveraging Android Intents/Deep Linking to pre-fill emergency messages for one-tap execution, or using server-side APIs (like Twilio) to bypass local OS restrictions entirely.
- **Offline Redundancy:** Demonstrating how local storage caching (SQLite/File system) securely holds the 10-second GPS breadcrumbs and evidence media until a network heartbeat or SMS capability is detected.
- **Ethical Boundaries:** Using a mock database for CNIC verification to prove the logic works without violating NADRA's real-world data privacy laws.

5. Expected Impact & Conclusion

Rahbar bridges the crucial gap between a tracking app and a proactive digital bodyguard. By addressing the realities of internet unreliability, street harassment, and public transit security in one unified platform, Rahbar offers a scalable, life-saving solution ready for implementation in modern smart-city infrastructures.